AdventureWorks Sales Analysis: Portfolio Summary

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The short version

I analysed the online sales of a fictional bicycle company (Microsoft's practice dataset, "AdventureWorks") from July 2005 to July 2008. The data covers 60,398 items sold in 27,659 orders to 18,484 customers, worth $29.4 million in revenue and $12.1 million in profit (a 41.1% margin). I worked through nine projects, from simple trends to customer groups, and finished with a one-page interactive dashboard. Along the way I checked my own conclusions, and some of them changed.

What I did

Beginner projects: what happened?

1. Sales over time. I added up revenue by year and month. Sales grew from $3.3M in the half-year of 2005 to $9.8M in 2007, while the profit margin stayed steady at about 41%. The years 2005 and 2008 are partial, so I only compared like with like.

2. Products. The best sellers were mountain and road bikes: ten models make up 93% of revenue, and the Mountain-200 model alone is 27%. From 2007, cheap accessories (items under $100) started selling. They are about 75-80% of the items sold but only 4-6% of revenue, which explains why the average order value fell from $3,224 in 2005 to $749 in 2008.

3. Countries. The United States (32%) and Australia (31%) bring in about 63% of revenue, and the UK, Germany, France and Canada share the rest. Profit margins were almost the same everywhere (about 41-42%), so the differences come from how much was sold, not how profitable it was.

Intermediate projects: who buys what, and when?

4. Customers. About half of customers (8,974) bought only cheap accessories: roughly $60 each and under 2% of revenue. The other half average $3,030 each and bring in 98% of revenue. Whether someone buys a bike is linked to things like how many children and cars they have. Once they do buy, higher-income customers spend more (about $4,100 or more above $90K income, against about $2,600 below $60K).

5. Product mix. Road bikes earn the most revenue (50%), and mountain bikes have the best margin (46%). Colours and sizes turned out to be tied to specific models, so I could not say that customers "prefer red", only that the red bikes are road models.

6. Seasonality. The day of the week makes no difference to sales, and December is the only month that is reliably strong. Looking at the company's financial year (July to June) instead of the calendar year revealed a drop of 18.5% in FY2007 that the calendar view had hidden, followed by growth of 186% in FY2008.

Advanced projects: customer groups, repeat buying, and a dashboard

7. Customer groups. I scored every customer on how recently they bought, how often, and how much they spent. About 12% of customers ("Champions") produce 38% of revenue, and a further 11% are big spenders due to buy again (35% of revenue). Only 255 big spenders look genuinely overdue, and they are the group worth a win-back offer. My first version called 2,361 customers "at risk", and I cut that to 255 after testing it.

8. Delivery and repeat buying. Every order shipped exactly 7 days after it was placed and was due exactly 12 days after, so the delivery data has no real information and I drew no conclusions from it. Instead I studied repeat buying. Nobody ordered twice before 1 July 2007, the same day accessories first appeared. Compared fairly, 45% of the oldest group of customers reordered within six months, then 36% of the middle group and 16% of the newest.

9. The dashboard. A one-page, interactive Excel dashboard with five headline numbers, a monthly sales trend, sales by country and by product line, filter buttons (slicers) for year, country and product line, and a box of key findings. It is available as a PDF.

What I learned about doing analysis honestly

- **Check that the totals add up.** Every new calculation was tied back to the $29.4M total.
- **Compare like with like.** Partial years, and groups watched for different lengths of time, gave misleading answers until I made the comparison fair.
- **Look at the data before trusting a column.** The delivery dates had no variation at all.
- **Averages can hide two different groups.** The average customer spend of $1,588 hid accessory-only buyers and bike buyers.
- **Test your own explanations.** I assumed early customers came back for accessories, but the data showed they came back for bikes. I also rebuilt my customer groups after a benchmark I had used turned out to be weaker than I thought.
- **Be clear about limits.** The patterns here are associations, not proof of cause.

Skills used

Microsoft Excel 2016: Tables, lookups (INDEX/MATCH, VLOOKUP), conditional sums and counts (SUMIFS, COUNTIFS), nested IF logic, PivotTables and PivotCharts, calculated fields, date and fiscal-calendar analysis, customer scoring (RFM), cohort analysis, slicers, and dashboard design.

Limits of this work

- This is a practice dataset, not a real company's records.
- The data has no product-category table, so "bike versus accessory" is approximated by price (under $100).
- The years 2005 and 2008 are partial, and July 2008 is almost empty.

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